

<page_start no = 1>

<sol_start id=1>

1. Convex lens

<sol_end>

<sol_start id=2>

2. Convex lens of 10 cm focal length has greater power.

<sol_end>

<sol_start id=3>

3. Focal length is define as the distance between the centre of the lens or mirror and its focal point.

$$u = -5 \text{ cm}$$

$$v = 10 \text{ cm}$$

Using lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{(-5)}$$

$$\frac{1}{f} \pm \frac{3}{10}$$

$$f = \frac{10}{3}$$

<sol_end>

<page_end>

<page_start no = 2>

<sol_start id=4>

4. Using lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Here v and u are the position of the image and object with respect to lens.

Magnification is the ratio of size of the image and the size of the object.

$$m = \frac{v}{u}$$

Hence image is inverted

$$-1 = \frac{v}{u}$$

$$v = -u$$

Given that image is formed at the distance from +50 from the lens and since the image is real.

$$v = +50 \text{ and } u = -50$$

$$\frac{1}{f} = \frac{1}{50} - \frac{(-1)}{50}$$

$$\frac{1}{f} = \frac{2}{50} = \frac{1}{25}$$

$$f = 25$$

$$\text{Power of the } P = \frac{1}{f}$$

$$P = \frac{1}{25} \times 10^{-2} \text{ m}^{-1}$$

$$P = \frac{100}{25} = 4\text{m}^{-1}$$

$$f = 1\text{m}^{-1} = 10$$

$$\text{So } P = 4D$$

<sol_end>

<page_end>